

Final Exam: Database Management Systems – SQL Section

Part A – Basic SQL (Solutions)

1. Write SQL statements to create the following table Students with constraints:

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    dob DATE,  
    department_id INT,  
    FOREIGN KEY (department_id) REFERENCES Departments(department_id)  
);
```

2. Insert two new rows into Students:

```
INSERT INTO Students (student_id, name, email, dob, department_id)  
VALUES (101, 'John Doe', 'john@example.com', '2002-06-15', 2),  
       (102, 'Jane Smith', 'jane@example.com', '2001-10-30', 1);
```

3. Retrieve all students born after 2002, sorted by name in descending order:

```
SELECT *  
FROM Students  
WHERE YEAR(dob) > 2002  
ORDER BY name DESC;
```

4. Display the total number of students in each department:

```
SELECT department_id, COUNT(*) AS total_students  
FROM Students  
GROUP BY department_id;
```

5. Write a query to delete all students from the Students table whose dob is before 2000-01-01:

```
DELETE FROM Students  
WHERE dob < '2000-01-01';
```